Energy Demand Forecasting & RL Load Scheduling

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ABSTRACT

This study presents an integrated framework for residential energy management that combines time-series forecasting with reinforcement learning-based load scheduling. The system employs Long Short-Term Memory (LSTM) and Temporal Convolutional Network (TCN) models to predict next-hour household electricity demand using historical consumption data. These predictions are subsequently utilized by a Q-learning agent to determine whether flexible appliances should be executed immediately or delayed, with the objective of reducing proxy energy cost.

An auxiliary Natural Language Processing (NLP) component is incorporated to enhance interpretability by classifying demand-related textual descriptions into increase, decrease, and stable categories. Due to the absence of explicit appliance scheduling data, the system operates within an offline simulation environment where appliance requests are synthetically generated.

Experimental results show that the LSTM model achieves the best forecasting performance, with a Mean Absolute Error (MAE) of 0.3148 and a Mean Absolute Percentage Error (MAPE) of 0.4108, outperforming both the TCN model (MAE = 0.3222, MAPE = 0.4519) and the linear regression baseline (MAE = 0.3790, MAPE = 0.5876). The reinforcement learning agent further demonstrates the ability to reduce proxy energy cost, achieving an average cost reduction of 1026.00 compared to the baseline scheduling policy. The NLP component achieves moderate performance, with an accuracy of 0.5507 and a Macro-F1 score of 0.4977, serving primarily as an interpretability layer.

While the proposed system is limited by dataset constraints and simulation assumptions, the results highlight the effectiveness of integrating predictive modeling and decision-making for intelligent energy management. The framework provides a foundation for future work involving real-world deployment, incorporation of appliance-level data, and integration of dynamic pricing signals. A Streamlit-based dashboard is also developed to visualize system outputs and support interactive analysis.

CCS Concepts

• Computing methodologies → Machine Learning; Reinforcement learning; • Applied computing → Smart grid; • Information Systems → Data analytics; • Computer systems organization → Embedded and cyber-physical systems

Keywords

Energy demand forecasting; Reinforcement learning; Load scheduling; Time-series prediction; LSTM; Temporal Convolutional Networks; Smart energy systems; Machine learning

1. INTRODUCTION

Electricity demand in residential environments exhibits significant temporal variability influenced by human behavior, appliance usage patterns, and daily routines. These fluctuations often lead to peak demand periods, which contribute to increased operational costs and inefficient energy utilization. As energy systems continue to evolve toward intelligent and adaptive infrastructures, there is an increasing need for solutions that not only predict future demand but also optimize consumption through informed decision-making.

Time-series forecasting techniques enable the estimation of future electricity demand based on historical observations. However, forecasting alone does not inherently improve system efficiency unless it is integrated with a mechanism capable of acting upon predicted values. Without such integration, predictive insights remain underutilized and fail to translate into practical energy savings.

This study proposes an integrated intelligent system that combines deep learning-based forecasting with reinforcement learning-based load scheduling. Specifically, Long Short-Term Memory (LSTM) and Temporal Convolutional Network (TCN) models are employed to predict next-hour household electricity demand. These predictions are subsequently utilized by a Q-learning agent to determine whether a flexible appliance should be executed immediately or delayed in order to reduce proxy energy cost.

The system is designed as an offline simulation prototype. Due to the absence of explicit schedulable appliance events in the dataset, flexible appliance requests are synthetically generated. The overall framework follows a forecast-then-schedule paradigm, where predictive outputs guide decision-making policies within a simulated environment. This approach allows the study to explore the interaction between prediction and control while maintaining feasibility within the constraints of available data.

2. PROBLEM DEFINITION AND MOTIVATION

Residential electricity consumption is inherently dynamic and often unpredictable, yet scheduling decisions are typically made without incorporating knowledge of future demand. This results in inefficient energy usage, particularly during peak periods when electricity demand is already high. Traditional scheduling approaches lack predictive awareness, limiting their ability to optimize consumption patterns effectively.

The primary objective of this study is to address this limitation by integrating forecasting and reinforcement learning into a unified framework. The forecasting component estimates future demand, while the reinforcement learning agent uses these predictions to determine optimal scheduling actions. By shifting flexible loads to

more favorable time periods, the system aims to reduce overall energy cost and improve efficiency.

The performance of the system is evaluated across multiple dimensions. Forecasting performance is assessed using Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE), which measure prediction accuracy. The auxiliary Natural Language Processing (NLP) component is evaluated using accuracy and Macro-F1 score, reflecting its classification performance. The reinforcement learning component is evaluated using reward trends, cost reduction relative to a baseline policy, and consistency across multiple training seeds.

Several constraints must be acknowledged. The dataset represents only a single household, which limits generalizability. Additionally, the absence of real appliance scheduling labels necessitates the use of a simulated environment. The system is therefore intended as an academic prototype rather than a production-ready solution.

3. DATASET AND PREPROCESSING

The system utilizes the UCI Individual Household Electric Power Consumption dataset, which contains minute-level electricity usage data collected over approximately four years. The dataset is publicly available from the UCI Machine Learning Repository (Hebrail & Berard, 2006). However, its representation of a single household introduces limitations in terms of diversity and generalizability.

The preprocessing pipeline transforms the raw dataset into a structured format suitable for modeling. Missing values are handled using forward-fill and backward-fill techniques to preserve temporal continuity. The original date and time attributes are combined into a unified datetime field, after which the data is resampled into hourly intervals. This aggregation reduces computational complexity while preserving meaningful temporal patterns. Feature scaling is applied using StandardScaler. The scaler is fitted on the training data and applied to validation and test sets to avoid data leakage.

Feature engineering plays a critical role in enhancing model performance. Temporal features such as hour of day, day of week, and month are extracted to capture periodic patterns in electricity consumption. A weekend indicator is included to distinguish between weekday and weekend behavior. Additionally, sinusoidal encodings are applied to represent cyclical time features, enabling models to better capture periodic trends.

The dataset is split chronologically into training, validation, and test sets using a 70–15–15 ratio. This chronological split prevents data leakage and ensures that evaluation results reflect realistic forecasting conditions, where future data is not available during training.

4. SYSTEM DESIGN

The proposed system integrates forecasting, NLP, and reinforcement learning components into a unified pipeline. Each component is designed to fulfill a specific role while contributing to the overall objective of demand-aware scheduling.

The forecasting module consists of three models: a linear regression baseline, an LSTM model, and a Temporal Convolutional Network (TCN). The linear regression model serves as a simple non-deep-learning baseline, providing a reference point for evaluating the effectiveness of more complex models. The LSTM model is selected due to its ability to capture temporal dependencies and sequential patterns in time-series data, making it well-suited for modeling electricity consumption trends. The TCN is included as a convolutional alternative that uses dilated convolutions to capture temporal patterns while enabling parallel computation.

An auxiliary NLP component is incorporated to enhance interpretability. This module converts structured energy context into textual descriptions and classifies them into categories such as increase, decrease, and stable using TF-IDF vectorization and logistic regression. While not central to prediction or scheduling, this component provides a human-readable layer that supports system understanding.

The reinforcement learning component is implemented using a Q-learning agent. This approach is chosen due to its simplicity, interpretability, and suitability for discrete action spaces. The agent operates within a simulated environment that uses predicted demand values and synthetically generated appliance requests. At each time step, the agent selects an action—either running the appliance immediately, delaying it, or taking no action if no appliance is present. To improve realism, constraints such as maximum delay were introduced for simulated appliances, preventing trivial policies such as always delaying execution.

The system compares the agent's decisions against a baseline policy that always executes appliances immediately. This comparison enables the evaluation of cost reduction achieved through learned scheduling behavior.

5. TRAINING AND RESPONSIBILITY

The system is implemented using PyTorch for deep learning models and scikit-learn for baseline and NLP components. Training configurations are defined in a centralized configuration file to ensure consistency across experiments.

The forecasting models are trained using the Adam optimizer with Mean Squared Error (MSE) loss. Training is conducted for a maximum of 20 epochs, with early stopping applied based on validation performance using a patience of 4 epochs. No learning rate scheduler is used. A fixed random seed (42) is applied to ensure reproducibility in forecasting experiments, while the reinforcement learning component is evaluated across three different seeds (42, 52, and 62) to assess stability.

All experiments are conducted on a desktop computer equipped with an AMD Ryzen 5 5600 processor (6 cores, 12 threads, 3.50 GHz) and 16 GB of RAM, running Windows 11 Home (64-bit). Training is performed on CPU without GPU acceleration. The development environment consists of Visual Studio Code with a Python virtual environment, and all experiments are executed locally.

6. EVALUATION AND RESULTS

All results are computed on a held-out test using a chronological split, ensuring that evaluation reflects realistic forecasting conditions.

6.1 Forecasting Performance

Table 1. Forecasting Results

Model	MAE	MAPE
Linear Regression	0.3790	0.5876
LSTM	0.3148	0.4108
TCN	0.3222	0.4519

The forecasting results indicate that the LSTM model achieves the best performance among all evaluated models, as evidenced by its lowest MAE and MAPE values. This performance advantage can be attributed to its ability to effectively model temporal dependencies and sequential patterns in electricity consumption data.

In contrast, the linear regression model exhibits significantly higher error values, reflecting its inability to capture non-linear relationships and temporal dynamics. The TCN model demonstrates competitive performance, outperforming the baseline but slightly underperforming compared to LSTM. This suggests that while convolution-based approaches can model temporal patterns, recurrent architectures may provide better performance for this specific dataset.

6.2 Ablation Analysis

Table 2. Ablation Results

Experiment	Observation
Baseline vs LSTM	LSTM significantly outperforms linear regression
LSTM vs TCN	LSTM performs slightly better than TCN

The ablation results confirm that model complexity plays a crucial role in forecasting performance. The substantial improvement of LSTM over the baseline highlights the importance of capturing temporal dependencies. The comparison between LSTM and TCN further demonstrates that while both models are effective, LSTM provides a slight advantage due to its memory-based architecture.

6.3 Reinforcement Learning Performance

Table 3. Reinforcement Learning Results

Metric	Value
Mean Final Reward	1372.67
Mean Cost Reduction	1026.00
Mean Success Rate	0.9996
Mean Baseline Cost	7180.15
Mean Scheduled Cost	6154.15

The reinforcement learning results demonstrate that the Q-learning agent can learn a scheduling policy that reduces proxy energy cost relative to the baseline approach. The reduction in scheduled cost compared to baseline cost indicates that delaying appliance usage during high-demand periods can lead to improved efficiency.

However, the extremely high success rate should be interpreted cautiously. This value is influenced by the structure of the simulated environment, where many time steps contain no active appliance requests. As a result, the agent frequently selects the *no_action* decision, which contributes to a high success rate.

6.4 NLP Performance

Table 4. NLP Component Values

Metric	Value
Accuracy	0.5507
Macro-F1	0.4977

The NLP component achieves moderate performance and primarily serves as an interpretability layer. Its role is to provide human-readable insights into demand trends rather than to drive predictive performance.

6.5 Error and Slice Analysis

Table 5. Slice Analysis Results

Slice Category	MAE
Weekday	0.299
Weekend	0.354

The slice analysis reveals that forecasting performance varies across temporal conditions. Errors are higher during peak demand periods, particularly in the early morning and evening. Weekend data also exhibit higher errors, reflecting more irregular consumption patterns.

6.6 Calibration

Calibration was not included due to scope constraints. Since the system performs regression-based forecasting rather than probabilistic prediction, calibration techniques were not applicable.

7. ETHICAL CONSIDERATIONS AND POLICY

The proposed system is intended strictly for academic and experimental use. Its design as an offline simulation, combined with the use of a single-household dataset, introduces several ethical considerations related to generalizability, fairness, privacy, and potential real-world deployment.

From a data governance and privacy perspective, the dataset used in this study does not contain personally identifiable information (PII), making it suitable for research purposes. However, real-world implementations of such systems would require continuous collection of household energy consumption data, which may indirectly reveal sensitive behavioral patterns, such as occupancy schedules and lifestyle habits. Therefore, any deployment must incorporate strict data protection mechanisms, including anonymization, secure data storage, and compliance with applicable data privacy regulations.

In terms of fairness and representativeness, the dataset represents only a single household, which introduces inherent bias. Energy consumption patterns vary significantly across households due to differences in demographics, appliance usage, and socioeconomic factors. As a result, the trained models may not generalize effectively to other users. This limitation highlights the need for more diverse datasets in future work to ensure equitable performance across different populations.

The use of a simulated reinforcement learning environment introduces additional considerations. Appliance requests are synthetically generated, and the reward function is based on proxy cost rather than real electricity pricing. While this allows for controlled experimentation, it may lead to discrepancies between simulated performance and real-world outcomes. Consequently, the results should not be interpreted as directly transferable to real-world energy systems without further validation.

From a user autonomy and control perspective, automated scheduling systems have the potential to override user preferences if not properly designed. In real-world applications, it is essential to incorporate user override mechanisms, customizable preferences, and transparent decision explanations to maintain user trust and acceptance.

Additionally, there is a risk of model misinterpretation and overreliance. Users or stakeholders may assume that the system guarantees optimal performance under all conditions, despite its limitations. This is particularly relevant during irregular demand patterns or sudden spikes, where the model may produce inaccurate predictions. Clear communication of system limitations and failure cases is therefore necessary.

Finally, considerations related to sustainability and societal impact must be acknowledged. While the system aims to improve energy efficiency, its effectiveness depends on accurate forecasting and realistic deployment conditions. Future work should explore integration with real-time pricing, renewable energy sources, and broader smart grid systems to maximize societal benefit.

In summary, while the proposed system demonstrates promising capabilities in a controlled setting, responsible deployment requires careful attention to privacy, fairness, transparency, and user autonomy. These considerations are critical for ensuring that intelligent energy management systems are both effective and ethically sound.

8. MODEL CARD SUMMARY

The proposed system is designed as an academic prototype that demonstrates the integration of time-series forecasting and reinforcement learning for residential energy management. It is intended for simulated smart-home environments and educational purposes.

The system is not intended for real-time deployment, safety-critical applications, or direct use in operational energy systems. Its performance is based on a single-household dataset and a simulated scheduling environment, which limits its generalizability and real-world applicability.

For deployment in real-world settings, additional infrastructure would be required. This includes IoT sensors or smart meters for real-time data collection, appliance-level monitoring systems to detect and control device usage, and access to dynamic electricity pricing signals. Furthermore, a real-time data processing pipeline and user-facing controls, such as manual overrides and preference settings, would be necessary to ensure usability and user acceptance.

The system has several known limitations. It performs poorly during sudden spikes in energy demand and irregular usage patterns, where actual consumption may significantly exceed predicted values. Additionally, the reinforcement learning component operates in a simulated environment and may not reflect real-world scheduling constraints or user behavior.

Users and stakeholders should exercise caution when interpreting results, as the system does not guarantee optimal performance under all conditions. Any real-world implementation must include validation on diverse datasets and incorporate safeguards for user control, privacy, and fairness.

9. REPRODUCTABILITY AND EXECUTION

The proposed system is designed to be fully reproducible using the provided source code and configuration files. The complete pipeline, including data preprocessing, model training, evaluation, NLP classification, and reinforcement learning simulation, can be executed in a consistent and deterministic manner.

To set up the environment, the required dependencies must first be installed using the provided requirements file:

pip install -r requirements.txt

It is recommended to create and activate a Python virtual environment prior to installation to ensure proper dependency management and avoid conflicts with existing packages.

The dataset used in this study is publicly available from the UCI Machine Learning Repository and can be obtained from the following link:

https://archive.ics.uci.edu/dataset/235/individual+household+electric+power+consumption

The raw dataset file is automatically retrieved by the helper script when the pipeline is executed, and it is saved to:

data/household_power_consumption.txt

Once the environment and dataset are properly configured, the entire pipeline can be executed using a single command:

python run.py

This command sequentially performs all stages of the system, including dataset retrieval, preprocessing, model training, evaluation, NLP classification, and reinforcement learning simulation.

For visualization and interactive analysis, a Streamlit-based dashboard is provided. The dashboard can be launched using:

streamlit run app.py

All generated outputs, including model predictions, evaluation metrics, and simulation results, are stored in the results/ and logs/ directories, ensuring transparency and reproducibility of experimental findings.

10. CONCLUSION

This study presented an integrated framework for residential energy management that combines deep learning-based demand forecasting with reinforcement learning-based load scheduling. By employing LSTM and TCN models for time-series prediction and a Q-learning agent for scheduling decisions, the system demonstrates how predictive insights can be effectively translated into actionable control strategies.

The results indicate that the LSTM model achieves the best forecasting performance, highlighting its effectiveness in capturing temporal dependencies in electricity consumption data. The reinforcement learning component further demonstrates the ability to reduce proxy energy cost by intelligently delaying flexible appliance usage during high-demand periods. Although the NLP component achieves moderate classification performance, it

contributes to system interpretability by providing human-readable descriptions of demand trends.

Despite these promising results, several limitations must be acknowledged. The use of a single-household dataset restricts the generalizability of the findings, and the absence of real appliance scheduling data necessitates a simulated environment. Additionally, the reinforcement learning results are influenced by the structure of the simulation, particularly the presence of time steps without active appliance requests.

Future work should focus on addressing these limitations by incorporating more diverse datasets, integrating real appliance-level data, and extending the system to operate in real-time environments. Enhancements such as dynamic pricing signals, user preference modeling, and more advanced reinforcement learning algorithms may further improve system performance and applicability.

In conclusion, the proposed system demonstrates the feasibility and potential benefits of combining forecasting and reinforcement learning for intelligent energy management. While currently limited to a simulation setting, the framework provides a strong foundation for future research and real-world implementation in smart energy systems.

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